Lesson 6.3 Guided Notes

Session 1 · Writing Repeated Factors as Powers

I can write repeated multiplication as a power and name its base and exponent. · 6.EE.A.1

Word / Spanish	Meaning — complete the blank	Example
power / potencia	A short way to write the same _____ multiplied again and again.	9^4
base / base	The number that is _____ as a factor.	In 9^4, the base is 9
exponent / exponente	The raised number that tells how many _____ the base is a factor.	In 9^4, the exponent is 4

Key idea

The _____ is the repeated factor.

The _____ tells how many times the base is used as a factor.

The exponent counts factors, not _____ signs.

Worked example

Write 2 × 2 × 2 × 2 as a power. Name the base and the exponent.

The repeated factor is _____. That is the base.

Count the factors: there are _____. That is the exponent.

Write the base with a raised exponent: 2^4.

Answer: 2^4, read “2 to the _____ power”

You try

1. Write each product as a power.
a. 6 × 6 × 6 = _____
b. 1/4 × 1/4 = _____

_____

2. Write 10^4 in expanded form. How many factors does it have?

_____

Sum it up: In 7^4, the 4 tells me ___ because ___.

_____

Session 2 · Finding the Value of a Power

I can find the value of a power by multiplying the base the number of times the exponent shows. · 6.EE.A.1

Word / Spanish	Meaning — complete the blank	Example
evaluate / evaluar	Find the one _____ an expression equals.	$2^3 = 8$
squared / al cuadrado	To the 2nd power; use the base as a factor _____ times.	$4^2 = 4 \times 4 = 16$
cubed / al cubo	To the 3rd power; use the base as a factor _____ times.	$4^3 = 4 \times 4 \times 4 = 64$

Key idea

To evaluate a power, write the _____ form and multiply one factor at a time.

$4^3 = 4 \times 4 \times 4 = 64$, not $4 \times 3 =$ _____.

Fraction and decimal bases work the _____ way: $(1/2)^2 = 1/4$.

Worked example

Find the value of 6^3.

Expand: $6^3 = 6 \times 6 \times$ _____.

Multiply the first two: $6 \times 6 =$ _____.

Multiply by the last 6: $36 \times 6 =$ _____.

Answer: $6^3 =$ _____, not $6 \times 3 = 18$

You try

1. Find the value of each power.
a. $9^2 =$ _____
b. $0.5^2 =$ _____

_____

2. A cube has edges of 3 cm. Its volume is 3^3 cm³. Find the volume.

_____

Sum it up: 2^4 is not the same as 2×4 because ___.

_____

Session 1 · Model, reason, and explain

Writing Repeated Factors as Powers · 6.EE.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare 3 × 3 × 3 × 3, 3 × 4, and 3 + 3 + 3 + 3. Write the repeated-factor expression as a power and explain the others.

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: multiplies base by exponent.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Finding the Value of a Power · 6.EE.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare 2^5 and 5^2 by expanding both. Does reversing base and exponent preserve value?

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: treats exponentiation as commutative multiplication.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?